

6	B	Net work done = change in kinetic energy Area under F-s graph = Net work done

Qn	Ans	Solution
		$\Delta KE = \text{Area under } F - s \text{ graph}$ $\frac{1}{2}mv^2 - \frac{1}{2}mu^2 = (8.0)(2.0) + \frac{1}{2}(1.0)(8.0) + \frac{1}{2}(1.0)(-4.0)$ $\frac{1}{2}(2.0)v^2 - \frac{1}{2}(2.0)(2.5)^2 = 16 + 4 - 2$ $v^2 = 24.25$ $v = 4.9 \text{ m s}^{-1}$
7	C	Consider FBD for car